UNIVERSITY OF CAMBRIDGE INTERNATIONAL EXAMINATIONS

GCE Ordinary Level

MARK SCHEME for the October/November 2008 question paper

5054 PHYSICS

5054/02 Paper 2 (Theory), maximum raw mark 75

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Do not accept fractions. No penalty for ⩾ 2 s. f. unless stated or for 1 s. f. where exactly correct. Only one unit and only one fraction penalty per question.

Section A

1 **(a)** diagram of two forces **and** resultant — B1
W / 6(N) **and** *T* / 8(N) marked on perp. forces **or** scale given — B1
10(.0 ±0.2) N — B1
35–39° from *T*/Y/horizontal or 51–55° from *W*/vertical **and** correct resultant — B1

(b) 10(.0) N **or** e.c.f. — B1 [5]

2 **(a)** 0.5(0) m — B1

(b) rotates/tilts/unbalanced/one side down/one side up — C1
rotates anticlockwise/down on left **or** head down **or** foot up — A1
(net) anticlockwise moment **or** moment on left > moment on right **or** weight/CM on left of pivot — B1 [4]

3 **(a)** *mgh* **or** *F* × *d* **or** 10 × 700 — C1
(–)7000 J — A1

(b) *Q/E/H* = *mc*Δ*T* **or** (Δ*T* =) 7000/(1) × 4200 — C1
1.7 **or** 1.67 **or** 5.5 — C1
8.9 °C e.c.f. **(a)** — A1 [5]

4 **(a)** **(i)** (*a* = Δ)*v*/*t* **or** 84/35 — C1
2.4 m/s² — A1

(ii) speed **and** time axes correct **and** labelled — B1
straight line of positive gradient through origin — B1
84 (m/s) **and** 35 (s) marked — B1

(b) **(i)** two arrows with forward force > backward force — B1

(ii) air/wind resistance **or** friction **or** drag — B1 [7]

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5 **(a)** Any **two** pairs – may be expressed in terms of the gas:

liquid **M1**	*molecules* **A1**	
dense(r)	close(r)/touching	
incompressible/volume fixed	close(r) **or** strong(er) forces	
fills bottom container	forces strong(er)	
expands less **when heated**	forces strong(er)	
more viscous/flows slower	forces strong(er)	
sound fast(er)	close(r) **or** strong(er) forces	
better **conductors** of heat	close(r)	M2
slower diffusion	close(r)	A2

(b) molecules **gain** speed/energy/heat **and** escape/leave cloth/break bonds **or** latent heat needed B1
fast(er)/high(er) (kinetic) energy molecules escape/evaporate B1
(average) speed / (kinetic) energy (of remainder) decreases
or temperature related to (average) energy/speed of molecules B1 [7]

6 **(a)** red B1

(b) **(i)** equal to B1

(ii) less than B1

(c) two correct refractions on Fig. 6.2 M1
no dispersion **and** ray ends close to P A1 [5]

7 **(a)** 12(.0) V B1

(b) top row: 4.6 **and** 0 B1
bottom row: square 1 = square 2 + square 3 **or** 9.2 B1
bottom row: 4.6 in squares 2 **and** 3 **cao** B1

(c) (*E*=) *QV* **or** *VIt* **or** 200 × 12 C1
2400 J **accept** 2370–2410 J e.c.f. A1 [6]

8 **(a)** fusion B1

(b) **(i)** mass decreases **or** product/nuclei/atoms less massive B1
mass becomes/converted to energy B1

(ii) $E = mc^2$ B1
$6.6 \times 10^{-29} \times (3.0 \times 10^8)^2$ C1
5.9×10^{-12} **or** 5.94×10^{-12} J A1 [6]

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Section B

9 (a) (i) Any **three** lines:
vibration of **cone/loudspeaker** B1
vibration of **air/particles** (molecules) B1
particles/molecules pass on vibrations/energy (to neighbours) B1
compressions **and** rarefactions
or longitudinal wave/movement of particle B1
(max 3)

(ii) loud – large amplitude/max displacement B1
low-pitched – frequency/no. of waves per sec M1
low frequency, small frequency, etc. (long wavelength 1/2) A1

(iii) (t =) d/s **or** 0.57/330 C1
0.0017 s A1

(iv) speed of sound greater in water/liquid **or** v.v. B1
less time taken in water/liquid **or** heard sooner/faster B1 [10]

(b) (i) $v = f\lambda$ **or** 200 seen C1
(λ =) v/f **or** 330/200 **or** 330/0.2 **or** 1650 (m) C1
1.6/1.65/1.7 m A1

(ii) attempt at compressions and rarefactions/longitudinal wave M1
correct wavelength marked A1 [5]

[Total: 15]

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10	**(a)**	**(i)**	at least 2 concentric, complete circles	B1	
			increasing gap	B1	
			at least 1 anticlockwise arrow **and** none incorrect	B1	
		(ii)	stronger **or** more lines **or** lines closer together **or** extends further	B1	[4]
	(b)	**(i)**	(R =) V/I **or** 6.0/8.0	C1	
			0.75 Ω	A1	
		(ii)	(Q =) It **or** 8.0 × 120 **or** 8.0 × 2	C1	
			960 C (16 C scores 1/2)	A1	[4]
	(c)	**(i)**	L→R **or** N→S	B1	
		(ii)	force (on wire) **or** wire bends/moves	M1	
			into page/perpendicular to field/away (from us)/LH rule quoted	A1	
		(iii)	force reverses **or** out of page **or** bends the other way e.c.f.	B1	[4]
		(iv)	accept first two marks on unlabelled diagram		
			(wire becomes) coil / armature /solenoid	B1	
			force/movement opposite on sides of coil **or** moment	B1	
			current reverses during rotation/due to commutator or split ring	B1	[3]

[Total: 15]

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11 (a) (*P* =) *VI* **or** 6.0 × 1.6 C1
9.6 W A1 [2]

(b) (i) filament/J releases electrons
or thermionic emission B1
attracted by +ve terminal/metal plate/K B1
electrons move/accelerate B1

(ii) otherwise **electrons** hit (air) molecules/particles/lose energy
or electrons deflected/don't hit screen/cause ionisation of air B1

(iii) electrons/charges/beam/ray deflected (by magnetic field) B1
few(er) electrons reach plate/K/+ve terminal/pass round circuit B1

(iv) current = 0 **or** no reading B1
electrons repelled by **or** not attracted to K
or K does not emit electrons B1 [8]

(c) (i) (dot/speck of light) moves so fast (that the eye sees it as a single line) **or**
timebase pulls it horizontally **or** voltage is constant/zero B1

(ii) (line/trace) displaced vertically M1
at uniform rate/speed **or** slowly A1
moves 3.0 divisions/3cm B1

(iii) screen not high enough **or** trace moves beyond edge of screen
or line moves 6cm / more than 4cm (vertically) **or** line can only move 4cm **or**
screen is only 4cm from middle to top B1 [5]

[Total: 15]

B1 Independent mark
C1 Compensation mark; given also if the answer is correct
M1 Method mark:
if not given, subsequent A marks are not awarded
A1 Answer mark.

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